MANISHA CHAND

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EDUCATION

San Francisco State University — B.S. Computer Science | San Francisco, CA | December 2026 | [GPA:3.57]

Coursework: *Deep Learning, Natural Language Technologies, Artificial Intelligence, Data Structures & Algorithms, Analysis of Algorithms, Operating Systems, Computer Architecture & Assembly, Software Development, Software Engineering, Intro to Database Systems, Physics(Mechanics & Electrodynamics), Calculus, Linear Algebra, Discrete Mathematics, Probability & Statistics, Numerical Analysis*

EXPERIENCE

Research lead — Autonome Research Labs | 2026

- Open-source research group shipping modular tooling for programming work and literature review.
- Coordinating limited compute infrastructure across 3 continents for competitions and experiments in materials and biosciences.
- CAD agent harness for laser cutting, 3d printing, and CNC.

Member of Technical Staff — RJLF LLP | [Jun 2025] – [Aug 2026]

- Built agentic pipelines to retrieve and analyze judicial transcripts and opinions. Citation accuracy and critical observability .
- General transcript PDF parser with markers and accumulators that aggregate summaries and transcript components for pipeline RAG consumption.
- Designed and deployed auth pipelines for multiple applications via existing Microsoft authentication and external MFA options.
- Developed Vue and Quasar frontend stack.
- Worked within the Microsoft ecosystem to build automations and reduce cost overhead with outsourced AI platforms.
- Met with attorneys weekly to discuss needs and use cases, adjusting strategy to meet demand.

OSS Contributor — AI Alliance | 2026

- Built model-update validation and checkpoint fingerprint checks for consortium training.
- Fixed contribution build targets and added a CI regression check.

Software Developer & Research Intern —_People's Solar Energy Fund | [Jan] – [May] 2026

- Gathered internal data on networking with member organizations and built a common tool for member orgs to find mutual support through the PSEF network
- Completed research on the philosophy of commoning and produced tangible technical pathways to support PSEF's goals.
- Created a DevOps path for PSEF to enhance their digital network and governance strategies.

Programming Tutor & Grader — City College of San Francisco | 2024

- Supported **~40 students** in introductory Python and Java one-on-one; graded assignments on correctness, structure, and edge cases.

PROJECTS

FourcastNet3 Himalayas Fine-Tuning | Python, PEFT/LoRA, multi-GPU | →

- Developing a regional adaptation pipeline for FourCastNet 3 targeting temperature and wind forecasts over Nepal and the Himalayas. Coordinating experiments on two DGX Spark systems, with ERA5 data preparation, elevation-aware bias correction, and evaluation against frozen-model baselines.

CodexMicro Driver Operating Systems | C, Linux, Kernel Module | →

- Developed a virtual device driver in C that simulates the OpenAI Codex Micro agent-control macropad, enabling interaction with an emulated device or open-source physical implementations.

Basic File System Operating Systems | C, Linux | →

- Lead programmer for free-space management and the directory layer of a FAT-based file system built by a five-person team: block allocation/release via chain splicing, path resolution, directory persistence, and the buffered I/O layer.

Stack machine Interpreter | Java | →

- Stack-based VM that loads compiled bytecode, resolves jump labels to instruction addresses, and executes programs with recursion, dynamic heap allocation, a verbose trace mode, and bounds/use-after-free/OOM exception handling.

Multithreaded Police Dispatch Analytics | C, POSIX threads | →

- Processed a **342 MB fixed-length-record dataset (500,000 records, 36 fields)** with a configurable pthread pool, batching 2,000 records per thread and guarding shared accumulators with per-scope mutexes; reported response-time medians and quartiles as ASCII box plots.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, C++, SQL, Bash **Frontend:** Vue 3, Quasar, Pinia, React, Vite

Backend & Data: Node.js, Express, Prisma, PostgreSQL, Elasticsearch, SQLite, Django **Infra:** Docker, Git, GitHub Actions, Azure, Linux

AI/ML: vLLM, Kernel Engineering, Inference, LangChain, multi-agent orchestration, RAG, token budgeting, LoRA fine-tuning (PEFT), Claude Code, pi, GRPO, enterprise automations with LLM and cron-scheduled tasks

Microsoft: Power Apps, Power Automate, Dataverse, SharePoint. **Spoken:** English, Nepali, Hindi (fluent)